

Poland

Office of Competition and Consumer Protection

In recent years, AI has revolutionized the way organizations operate, providing solutions to problems that were once deemed very complicated or even unsolvable. The Office of Competition and Consumer Protection ('UOKiK') has implemented several AI tools to detect anticompetitive practices, enforce the law, and improve work processes. These tools have proven to be incredibly beneficial, helping the Office to operate more efficiently, effectively, and with greater accuracy. UOKiK has been able to increase the quality of its work overall and streamline its operations with the use of AI. Below is a description of most relevant AI tools implemented by the Office and their impact on the organization.

I. Bid Rigging Detection

In the area of competition protection, the UOKiK is developing a method for the automated detection of collusive bidding with the use of statistical tools. In recent years, we have conducted activities that result in analyses of data on large infrastructure projects (road and railway projects). One set contains data provided by the General Directorate for National Roads and Motorways. The database contains information on 3174 offers submitted. The second set contains data provided by the Polish State Railways. The database contains information on 1986 submitted offers.

On the basis of this data, we have calculated measures allowing to characterize the structure of the market and its dynamics, *i.e.*, measures of concentration, HHI, in general and regional terms. The analysis of the market structure does not show any alarming trend, the market is not dominated by one company or by a group of companies. Market concentration neither increases nor decreases appreciably. Its volatility is considerable, which is common in bidding markets where winning one large tender can lead to large changes in market share.

The first measure that was analyzed was the convergence of the price with the estimated value--it is considered that the price should be lower than the estimated value determined before the announcement of the tender by its organizer. If it is not, it may be the result of inflating the price by non-competitive behavior of bidders. The price-to-estimate ratio, measured as the ratio of the winning price to the estimate, was calculated. This measure exceeds 1 if the most advantageous price exceeds the value of the order estimated by the organizer.

Next, the volatility index was calculated. This indicator is determined as a relative standard deviation, *i.e.*, the standard deviation of prices divided by the average of the prices of bids submitted in a given tender (only tenders with more than 2 bids were taken into account). Then, procedures characterized by particularly high or low price volatility were identified.

The asymmetry of price distribution in individual tenders was also analyzed using the skewness index. Positive skewness means that the majority of offers are above average. This may indicate artificially expensive losing bids--the bids that are supposed to lose are set at high prices to make the winning bid seem more attractive against them. Another possible explanation is that the winning bid was underestimated, *e.g.*, as a result of a mistake or to gain needed experience.

Suspicious may also be raised by a particularly large gap between the first (best) and the second-best offer. To take into account the specificity of the market, the range index was calculated by dividing the range between the first and the second offer by the standard deviation of the offers excluding the first. This measure is intended to detect a situation in which the best price differs significantly from the others, whose values are rather concentrated.

It was also examined whether reciprocity can be observed between the companies whose bids were withdrawn and those which won the tender as a result (A withdraws and as a result B wins, then at some later date the situation is reversed - B withdraws, allowing A to win with a higher price).

In addition to situations of direct reciprocity between winners and withdrawers, more complex arrangements should also be considered. When firms strategically withdraw bids in multiple tenders, a deal is possible such that firm A withdraws in tender one, firm B wins that tender, withdraws in tender two, which is won by firm C, which in turn withdraws in tender three, so that company A wins. This situation can be called a cycle. Cycles can also be longer than three elements.

However, it must be taken into account that if these companies participate in tenders not on their own, but in consortia that are not permanent (in individual tenders, a given company participates in consortia of various composition), then the implementation of such an agreement would be much more difficult, and therefore there are doubts as to whether this is intentional.

II. Preliminary assessment of the risk of tender collusion

Regardless of the works described above, UOKiK conducts research and development work on the implementation of mechanisms for a preliminary assessment of the risk of tender collusion. Although we have a prototype tool that allows initial data analysis, the basic problem is the availability of high-quality data on the public procurement market. In order to develop appropriate databases, UOKiK cooperates with the Public Procurement Office, working on the use of data collected under the eProcurement (eZamówienia) system.

III. ARBUZ—AI-powered assistant detecting abusive contract clauses

Employees of UOKiK, react on received complaints or take action on the basis of market research, media reports and information from other sources. Many investigations are precedent-setting, requiring a multidimensional analysis of a particular problem. However, some activities are rather repetitive and laborious at the same time. Such tasks include the analysis of standard contracts in order to identify abusive clauses. UOKiK is obliged to ensure that contracts used in mass trade do not contain provisions detrimental to the interests of non-professional users. The search for such provisions consists in reading multipage standard contracts and then assessing which parts of them may infringe consumer rights. It is a time-consuming process. For this reason, UOKiK carried out an EU-funded project ‘ARBUZ,’ aimed at automating the process of analyzing standard contracts.

Employees involved in detecting illegal clauses can log into ARBUZ and start searching for model contracts. Already at this stage, it uses AI to assess whether the document found falls into this category. The program can also be fed with contracts received along with consumer complaints or obtained from other sources, for example, submitted by the investigated company at the request of UOKiK. ARBUZ is equipped with an OCR system, which allows you to work not only on editable documents, but also on pdfs, graphic files and scans in various formats.

ARBUZ reviews the contract selected by an employee. Using the slider, you can set the sensitivity with which the analysis is to be performed—e.g., 5, 10 or 20 illegal provisions are to be found. The system compares the contract uploaded manually or found with the help of the crawler, with clauses in the database. In two windows in the interface you can see the provisions which according to the robot may be unfair. In the first window the user sees the contract with automatically found clauses marked in color, and in the other, the same clauses with information on how likely they are, according to the system, to be unfair. ARBUZ does not just make a simple comparison

of the provisions of contracts with database entries but uses intelligent algorithms to recognize the meaning of complex sentences written in legal language. It works at the level of an intelligent assistant--it is more than a tool but at the same time it is not allowed to make independent decisions.

At the end of the process, however, there is a human case handler who decides how to deal with the results of the ARBUZ's work. First of all, he or she may recommend bringing charges against the entrepreneur or sending a letter requesting a change in the standard contract.

IV. Dark Patterns

One of the current online threats, against which practical standards of combating have not yet been developed, are dark patterns--user interface design choices that benefit an online service by coercing, steering, or deceiving consumers into making unintended and potentially harmful decisions.

UOKiK has initiated an EU-funded project to develop a methodology for conducting proceedings on dark patterns and explore AI deployment opportunities for consumer protection authorities through the delivery of a Proof of Concept of an AI-powered tool for detecting dark patterns as well as the research on the latest AI and machine learning trends. Websites of online stores and other online service providers will be screened and specific online market sectors with the highest dark patterns occurrence will be selected for further analysis.

Next, explanatory proceeding will be initiated on the basis of the findings of an open-source intelligence, which will also include neuromarketing tests will be carried out to examine neurobiological human reactions while being exposed to dark patterns. In the course of administrative proceedings, UOKiK plans to organize an IT analysis of websites, delivered by external experts, and conduct a consumer survey about online purchase experience and the influence of certain dark patterns.

In the next stage, the results of IT website analysis, consumer survey, and neuromarketing tests will be used as input data in the Proof-of-Concept process of a dark patterns AI-detector. The project team will also prepare the main assumptions of the AI tool and an open idea competition will be organized. The contest formula stimulates innovation and allows for the selection of the best contractor on the basis of their real competences rather than experience or market position.

The results of the project will be presented in the form of guidelines for conducting proceedings on dark patterns and a white paper on practical, legal, and ethical aspects of using AI by a public body enforcing consumer law.